

Stokes equation (simplified form)

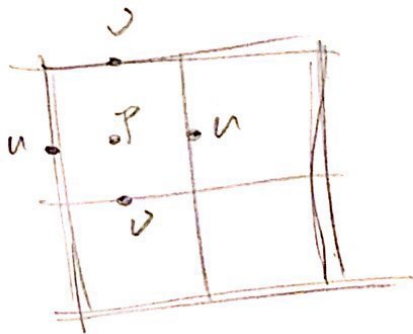
$$\begin{cases} -\Delta \vec{u} + \nabla p = \vec{f} \\ -\nabla \cdot \vec{u} = g \end{cases} \quad \text{in } \Omega$$

$$\text{div } \vec{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \quad \text{in 2-dimension } \Omega.$$

$$\vec{u} = (u, v), \quad \vec{f} = (f_1, f_2)$$

$$\begin{cases} -\Delta u + p_x = f_1 \\ -\Delta v + p_y = f_2 \\ u_x + v_y = g \end{cases}$$

[1.] MAC: Marker and Cell.



① in order to satisfy inf-sup condition

② for FDM to solve Stokes equation

* nothing new except the structure. all the rest are the same as Finite Difference Method. e.g. dealing with Laplacian operator and divergence operator.

[2.] DGS: Distributive Gauss-Seidel.

① DGS is a smoother used in the multigrid method for solving Stokes equation by MAC structure

② Why not standard relaxation, i.e. G-S relaxation?

the discrete structure:
$$\begin{pmatrix} A & B^T \\ B & 0 \end{pmatrix} \begin{pmatrix} \bar{u}_h \\ p_h \end{pmatrix} = \begin{pmatrix} \bar{f}_h \\ 0 \end{pmatrix}$$

this guy is not diagonally dominant
zero block hampers the relaxation.

$$\boxed{Lx = b}$$

idea of DGS:

our principle operator: $L = \begin{pmatrix} A & B^T \\ B & 0 \end{pmatrix}$

let $M = \begin{pmatrix} I & B^T \\ 0 & -BB^T \end{pmatrix}$

s.t. $LM = \begin{pmatrix} A & \underbrace{AB^T - B^T BB^T}_{BB^T} \\ B & BB^T \end{pmatrix} \approx 0$ in the sense of $\begin{cases} \text{① small} \\ \text{② low rank.} \end{cases}$ ✓

can be verified that $\text{rank}(AB^T - B^T BB^T) \leq n^2 - (n-1)^2$
comparing with the rank of n^2 , $= O(n^2)$
 $n^2 - (n-1)^2 = (n+n-1)(n-n+1) = 2n-1 = O(n)$ relatively small.

$$\tilde{L}\tilde{M} = \begin{pmatrix} A & 0 \\ B & BB^T \end{pmatrix} \approx LM$$

↑
approximation

$$\therefore L^{-1} = M(LM)^{-1} \approx M(\tilde{L}\tilde{M})^{-1}$$

So, the iterative method (residual-correction)

$$x^{k+1} = x^k + \underbrace{M(\tilde{L}\tilde{M})^{-1}}_{\substack{\uparrow \\ \text{estimate } L^{-1}}} \underbrace{(b - Lx^k)}_{\text{residual}}$$

correction

update form.

basically in Multigrid Method, we prefer solving residual equation. Here using DGS is the same.

i.e. we use $L^{-1} \approx M(\tilde{L}\tilde{M})^{-1}$ so $L \approx (\tilde{L}\tilde{M}) \cdot M^{-1}$

$$\text{so } \boxed{Le = r} \longrightarrow (\tilde{L}\tilde{M}) \cdot M^{-1}e = r \quad \tilde{e} := M^{-1}e$$

$$\tilde{L}\tilde{M} \cdot \tilde{e} = r$$

$$A_p := BB^T$$

$$\boxed{\begin{pmatrix} A & 0 \\ B & A_p \end{pmatrix} \tilde{e} = r}$$

solve this.

got $\tilde{e}$, the desired $e = M\tilde{e}$

← this is why called "distributive"

DGS details: $\hat{A} \sim A, \hat{A}_p \sim A_p = BB^T$

① relax the momentum equation.

$$u^{k+\frac{1}{2}} = u^k + \underbrace{\underbrace{\hat{A}^{-1}}_{\text{estimate } \hat{A}^{-1}}} \underbrace{(f - Au^k - B^T p^k)}_{\text{residual}} \quad \text{correction.}$$

update. however, this is not $u^k + e_n$. this is $u^k + \tilde{e}_n$, that's why we denote it by $u^{k+\frac{1}{2}}$ instead of u^{k+1} . this is an intermediate term for distribution later.

② relax the mass equation.

$$e_p = \underbrace{\hat{A}_p^{-1}}_{\text{estimate } \hat{A}_p^{-1}} (g - Bu^{k+\frac{1}{2}})$$

residual, here we use new $u^{k+\frac{1}{2}}$ instead of u^k

③ distribute corrections to original variables.

$$\begin{cases} u^{k+1} = \underline{u^{k+\frac{1}{2}}} + \underline{B^T e_p} \\ p^{k+1} = p^k - \underline{BB^T e_p} \end{cases} \quad \text{as } M\tilde{e} = e$$

$$M = \begin{pmatrix} I & B^T \\ 0 & -BB^T \end{pmatrix}$$

unrelated to e_n
so update directly.